Outline

- Inter-observational Distances
- Partition-based Clustering
- Hierarchical Clustering
- Diagnostics and Optimizing Number of Clusters
- Density-based Clustering

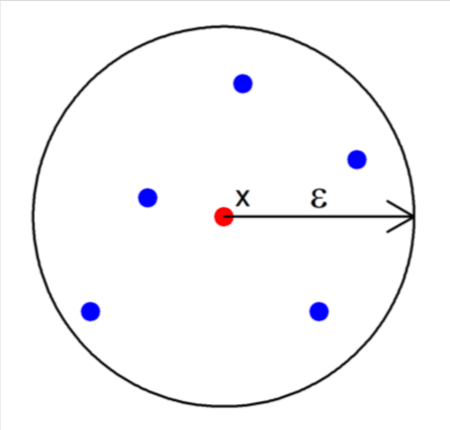
DBSCAN: Density-based Clustering Algorithm

- Unlike previous clustering algorithms, DBSCAN
  - Can find any shape of clusters
  - Identifies observations that do not belong to clusters as outliers
  - Does not require specifying the number of clusters
  - Can be used for predicting cluster membership for new data

DBSCAN: Density-based Clustering Algorithm

The DBSCAN algorithm identifies dense regions of observations. However, we need to specify two parameters for the algorithm:

1. ϵ: the radius of a neighborhood around an observation
2. $MinPts$: the minimum number of points within an ϵ radius of an observation to be considered a "core" point.



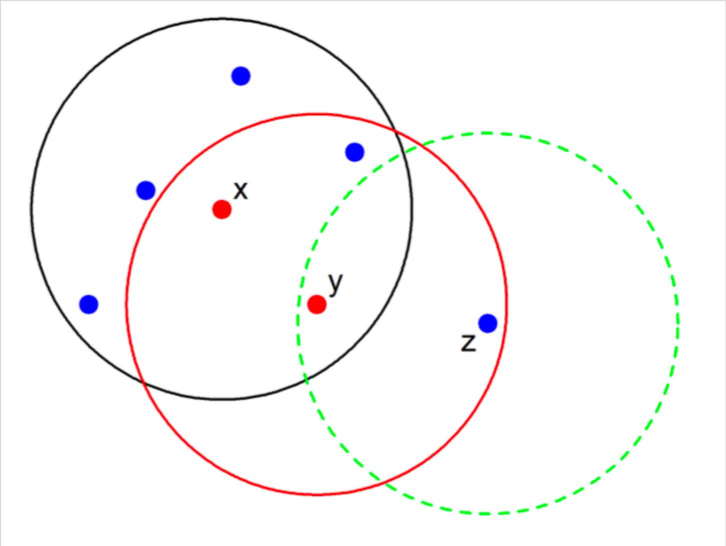
Example of core observation with $MinPts = 6$

DBSCAN: Density-based Clustering Algorithm

There are three types of points:

- **Core points:** observations with $MinPts$ total observations within an ϵ radius
- **Border points:** observations that are not core points, but are within ϵ of a core point
- **Noise points:** everything else

In this example, x is a core point, y is a border point, and z is a noise point.



DBSCAN: Density-based Clustering Algorithm

Two terms:

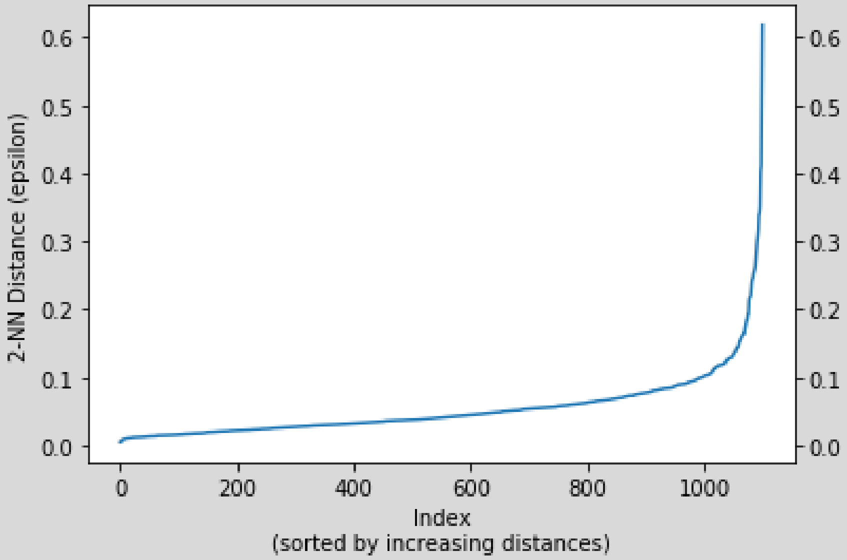
- **Density-reachable:** Point A is density-reachable from point B is there is a set of core points leading from B to A.
- **Density-connected:** Two points A and B are density-connected if there is a core point C such that both A and B are density-reachable from C.

Therefore, a density-based cluster is defined as a group of density-connected points.

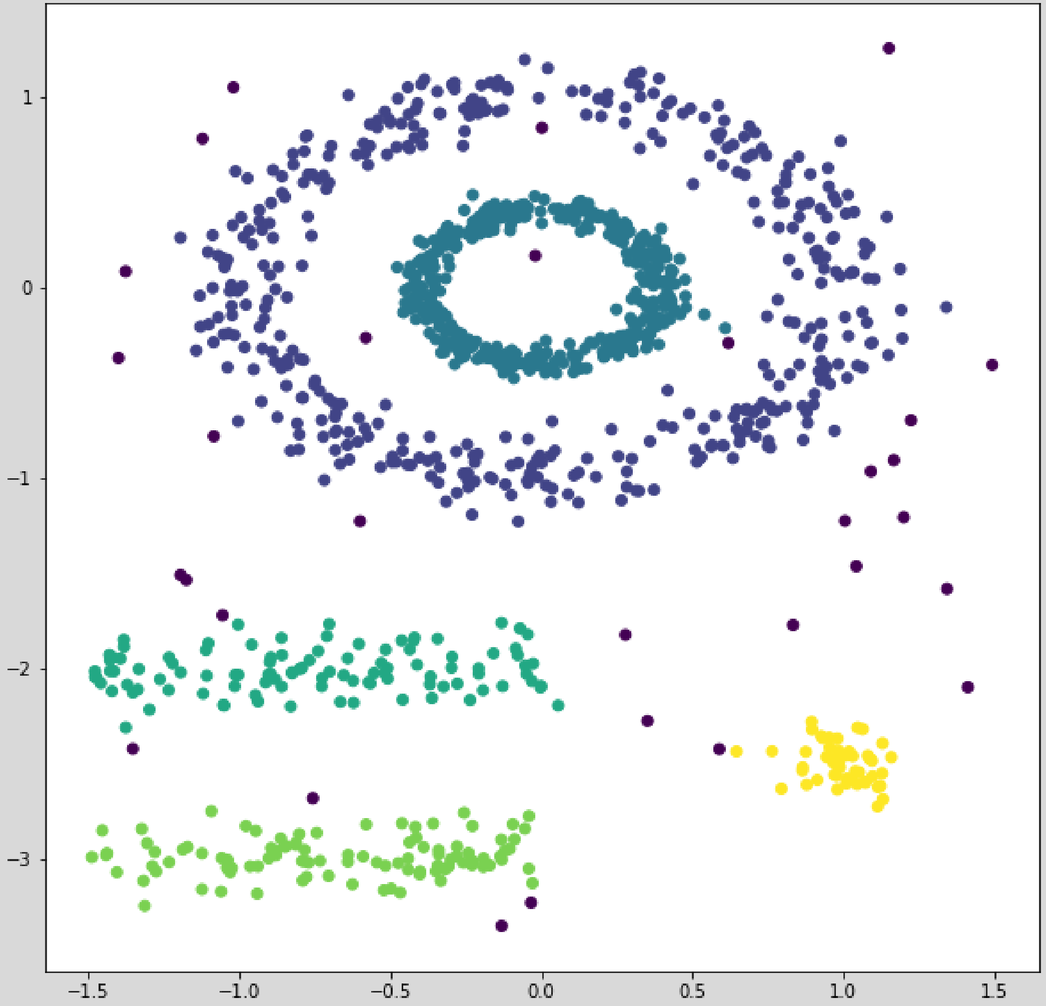
DBSCAN: Density-based Clustering Algorithm

Choosing ϵ:

- Compute the k-th nearest neighbor distance for each point
- Plot the distances in sorted order
- Look for a bend (the "knee") in the plot and use the distance at the knee as the choice of ϵ.



DBSCAN: Density-based Clustering Algorithm



Questions?